7.1

$$f(x) = x^3 + 6x^2 - 3x - 5$$

$$f'(x) = 3x^2 + 12x - 3$$

$$f''(x) = 6x + 12$$

$$3x^2 + 12x - 3 = 0$$

$$3(x^2 + 4 - 1) = 0$$

$$\frac{-4 \pm \sqrt{16+4}}{2} = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$$

$6(-2+\sqrt{5}) = -12 + 6\sqrt{5} \geq 0$, so this is a minimum

$6(-2-\sqrt{5}) = -12 - 6\sqrt{5} < 0$, so this is a maximum

There are no stationary points because we do not have a point where second derivative is 0.

7.2

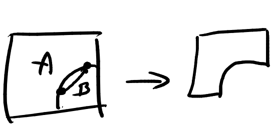
$$\theta_{i+1} = \theta_i - \gamma_i \nabla L(\theta_i))^T$$

7.3

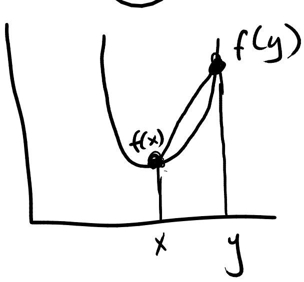
a) True

b) False

c) False



7.4



line connecting f(x) and f(y) has to always be above or equal to function values evaluated at points between x and y,
$f(\theta x + (1-\theta)y)$

7.4

a) $+$ $g(\theta x + (1-\theta)y) \le \theta g(x) + (1-\theta) g(y)$

$f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta) f(y)$

$g(\theta x + (1-\theta)y) + f(\theta x + (1-\theta)y) \le \theta g(x) + (1-\theta) g(y) + \theta f(x) + (1-\theta) f(y)$

So the resulting sum of two convex functions would also be convex.

b) False. Choosing $f(x) = x^2$, $g(x) = 3x^2$

We get $f(x) - g(x) = x^2 - 3x^2 = -2x^2$ which is not convex

c) False. if we have $f(x) = x^2$ and $g(x) = x^2 - 10$

Multiplying $f(x)g(x)$ gives $x^4 - 10x^2$ which is not convex

d) A convex function that is defined over all $\mathbb{R}$ is not going to have a maximum unless it is defined to be constant

If $f(x) = c$ and $g(x) = d$, then $\max(f(x), g(x))$ is going to be a convex function as well.

7.5

$$\max \; p^T x + \varepsilon$$

subject to

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \varepsilon \end{bmatrix} \le \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$$

7.6

$$\max_{x \in \mathbb{R}^5} \; -[33 \; 85 \; -18]\, x$$

subject to

$$-\begin{bmatrix} 5 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 & 2 & -2 & 0 & 0 \\ 2 & -4 & 1 & -1 & 1 \end{bmatrix} x = 0$$

$$x \ge 0$$

(7.7)

$$c = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} \quad Q = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

$$\max_{x \in \mathbb{R}^4} (c + A^T \lambda)^T A Q^{-1} (c + A^T \lambda) - \lambda^T b$$

subject to $\lambda \geq 0$

(7.8)

$$L(w, \lambda) = \frac{1}{2} w^T w - \lambda (w^T x - 1)$$

$$\frac{\partial L}{\partial w} = w - \lambda x \qquad \begin{aligned} 0 &= w - \lambda x \\ w &= \lambda x \end{aligned}$$

$$\frac{1}{2} (\lambda x)^T (\lambda x) - \lambda (\lambda (x^T) x - 1)$$

$$= \frac{1}{2} \lambda^2 x^T x - \lambda^2 x^T x + \lambda$$

$$= -\frac{1}{2} \lambda^2 x^T x + \lambda$$

(7.9)

$$f^* = \sup_x \left(y^T x - \sum_d x_d \log x_d \right)$$

$$\frac{\partial f^*}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\sum_d y_d x_d - \sum_d x_d \log x_d \right]$$

$$= y_j - \left(\log x_j + x_j \frac{1}{x_j} \right) = y_j - \log x_j - 1$$

$$y_j - \log x_j - 1 = 0$$

$$\log x_j = y_j - 1$$

$$x_j = e^{y_j - 1}$$

$$f^*(y) = \sum_d y_d e^{y_d - 1} - \sum_d e^{y_d - 1} \log(e^{y_d - 1})$$

$$= \sum_d y_d e^{y_d - 1} - \sum_d e^{y_d - 1}(y_d - 1)$$

$$= \sum_d y_d e^{y_d - 1} - e^{y_d - 1}(y_d - 1)$$

$$= \sum_d e^{y_d - 1}(y_d - (y_d - 1)) = \sum_d e^{y_d - 1}(1) = \sum_d e^{y_d - 1}$$

7.10

$$f(x) = \frac{1}{2} x^T A x + b^T x + c$$

$$f^*(s) = \sup_x \left(s^T x - \left(\frac{1}{2} x^T A x + b^T x + c \right) \right)$$

$$\frac{\partial f^*}{\partial x} = s - Ax - b$$

$$s - Ax - b = 0$$

$$(A^{-1})(s - b) = x$$

$$f^*(s) = s^T(A^{-1}(s-b)) - \left(\frac{1}{2}((A^{-1})(s-b))^T A (A^{-1}(s-b)) + b^T(A^{-1}(s-b)) + c \right)$$

$$= \underline{s^T(A^{-1}(s-b))} - \frac{1}{2}(s-b)^T A^{-1}(s-b) - \underline{b^T A^{-1}(s-b)} - c$$

$$= (s-b)^T A^{-1}(s-b) - \frac{1}{2}(s-b)^T A^{-1}(s-b) - c$$

7.11

$$L(\alpha) = \max\{0, 1-\alpha\}$$

$$L^*(\beta) = \sup_{\alpha \in \mathbb{R}} \left(\beta\alpha - \max\{0, 1-\alpha\}\right)$$

$$= \sup_{\alpha \in \mathbb{R}} \begin{Bmatrix} \beta\alpha & \text{if } \alpha \geq 1 \\ \beta\alpha - (1-\alpha) & \text{if } \alpha < 1 \end{Bmatrix} = \sup_{\alpha \in \mathbb{R}} \begin{cases} \beta\alpha & \text{if } \alpha \geq 1 \\ (\beta+1)\alpha - 1 & \text{if } \alpha < 1 \end{cases}$$

Now we need to plug in diffent β values and find what the supremum is.

$\beta > 0$:

$$L^*(\beta) = \sup_{\alpha \in \mathbb{R}} \begin{cases} +\infty & \text{if } \alpha \geq 1 \\ \beta & \text{if } \alpha < 1 \end{cases} = +\infty$$

$-1 \leq \beta \leq 0$:

$$L^*(\beta) = \sup_{\alpha \in \mathbb{R}} \begin{cases} \beta & \text{if } \alpha \geq 1 \\ \beta & \text{if } \alpha < 1 \end{cases} = \beta$$

$\beta < -1$:

$$L^*(\beta) = \sup_{\alpha \in \mathbb{R}} \begin{cases} \beta & \text{if } \alpha > 1 \\ +\infty & \text{if } \alpha < 1 \end{cases} = +\infty$$

$$L^*(\beta) = \begin{cases} \beta & \text{if } -1 \leq \beta \leq 0 \\ +\infty & \text{if not } -1 \leq \beta \leq 0 \end{cases}$$

$$f(\beta) = L^*(\beta) + \frac{\gamma}{2}\beta^2$$

$$f^*(\alpha) = \sup_{\beta \in \mathbb{R}} \alpha\beta - L^*(\beta) - \frac{\gamma}{2}\beta^2 = \sup_{\beta \in [-1,0]} \alpha\beta - \beta - \frac{\gamma}{2}\beta^2$$

$$= \sup_{\beta \in [-1,0]} -\frac{y}{2}\beta^2 + \alpha\beta - \beta$$ quadratic

$$\frac{\partial}{\partial \beta}\left(-\frac{y}{2}\beta^2 + \alpha\beta - \beta\right) = -y\beta + \alpha - 1$$

$$-y\beta + \alpha - 1 = 0$$

$$\beta^* = \frac{\alpha - 1}{y}$$ maximum

We have $-1 \le \beta \le 0$, plug in $\frac{\alpha-1}{y}$ for β^*

$-1 \le \frac{\alpha-1}{y} \le 0$, so $1 - y \le \alpha \le 1$

if $1-y \le \alpha \le 1$:

$$f^*(\alpha) = -\frac{y}{2}\left(\frac{\alpha-1}{y}\right)^2 + \alpha\left(\frac{\alpha-1}{y}\right) - \left(\frac{\alpha-1}{y}\right) =$$

$$= -\frac{y}{2}\left(\frac{\alpha-1}{y}\right)^2 + \left(\frac{\alpha-1}{y}\right)(\alpha-1) =$$

$$= -\frac{y}{2}\frac{(\alpha-1)^2}{y^2} + \frac{(\alpha-1)^2}{y}$$

$$= -\frac{(\alpha-1)^2}{2y} + \frac{(\alpha-1)^2}{y} = \frac{(\alpha-1)^2}{2y}$$

if $\alpha < 1 - y$: (if $\beta^* < -1$)

$$f^*(\alpha) = -\frac{y}{2}(-1)^2 - \alpha + 1 = -\frac{y}{2} - \alpha + 1$$

if $a > 1$: (if $\beta^* > 0$)

$$f^*(\alpha) = -\frac{y}{2}0^2 + \alpha \cdot 0 - 0 = 0$$

$$f^*(\alpha) = \begin{cases} -\frac{y}{2} - \alpha + 1 & \text{if } a < 1-y \\ 0 & \text{if } a > 1 \\ \frac{(\alpha-1)^2}{2y} & \text{if } 1-y \le \alpha \le 1 \end{cases}$$